A Mechanical Analysis of Two-Axis Clinostats in Producing Time Averaged Microgravity

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1 Introduction

Research on how biological systems respond to microgravity conditions is essential for advancing the field of space exploration. Since microgravity cannot be perfectly replicated on Earth due to the presence of Earth's large gravitational field, it is crucial to analyze the effectiveness and limitations of different experimental methods designed to mimic the behavior of microgravity environments.

Clinostats are devices used to replicate microgravity through the continuous reorientation of the direction of acceleration felt by the biological sample. It is critical to note that the sample still feels the full magnitude of our gravitational acceleration vector, but the inertial contributions from our rotating reference frames time-average out the net direction of acceleration felt by the sample. For the case of the two-axis Clinostat, this is accomplished by rotating the sample along two rotating frames with chosen angular velocities.

Throughout this paper, we will be evaluating the theoretical background behind the replication of microgravity conditions by the two-axis Clinostat through an analysis of the spatial distribution of our acceleration vector and the time dependence of the averaged microgravity.

2 Modeling Assumptions

The following assumptions were made in order to simplify our model:

- Our particle is constrained to a fixed radius away from the center of rotation.
- Mechanical vibrations produced from the motors and friction from the rotating frames are negligible.

Without the constraint of a fixed radius, accounting for the fluid dynamics of the sample culture and Coriolis forces generated by the sample's motion becomes increasingly complex. This is also not an unreasonable assumption since we already expect a radial dependence in the effectiveness of microgravity conditions, so constraining container size should already be accounted for in experiments performed using the Clinostat. The assumption of mechanical vibrations can be assumed to be negligible given that the Clinostat is properly constructed with an emphasis of smooth motors. If a strong mechanical vibration is present, research has shown that cell division will be slowed, which makes experimental results difficult to interpret.

3 Clinostat Dynamics

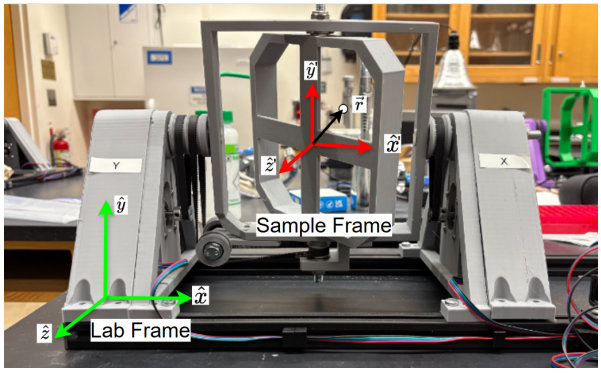


Figure 3.1: Diagram of Clinostat reference frames

`Figure 3.1` shows the reference frames we have constructed to describe the sample's motion in the Clinostat. The lab frame is our stationary frame, while the sample frame is our non-inertial reference frame. This allows us to describe the inertial contributions to the sample's acceleration in the lab frame using the following effective acceleration equation:

$$\vec{g}_{\text{eff, lab}}(\vec{r},t) = \vec{g}_0 - \dot{\vec{\omega}} \times \vec{r} - \vec{\omega} \times (\vec{\omega} \times \vec{r}) \tag{3.1}$$

Where $\vec{g}_{\text{eff, lab}}$ is the effective gravitational acceleration as observed in the stationary lab frame, $\vec{g}_0 = [0,-g,0]$, is the acceleration due to Earth's gravitational field, $\dot{\vec{\omega}} \times \vec{r}$ is our tangential or Eulerian Acceleration, and $\vec{\omega} \times (\vec{\omega} \times \vec{r})$ is the centrifugal acceleration. In order to simplify the derivation we are going to only pay attention to inertial terms, since we are going to have to transform $\vec{g}_0$ later in order to achieve a full effective acceleration equation in the sample frame.

3.1 Inertial Contributions as Observed in the Lab Frame

To work with this equation, we first need to define the angular velocity vector $\vec{\omega}$, which we derive using rotation matrices used to describe the rotation of the sample frame in the Clinostat.

The rotation matrix describing the clinostat motion is given by the composition

$$R(t) = R_x(\omega_x t) R_y(\omega_y t)$$

Where our rotation matrices are

$$R_x(\omega_x t) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\omega_x t) & -\sin(\omega_x t) \\ 0 & \sin(\omega_x t) & \cos(\omega_x t) \end{bmatrix} \quad R_y(\omega_y t) = \begin{bmatrix} \cos(\omega_y t) & 0 & \sin(\omega_y t) \\ 0 & 1 & 0 \\ -\sin(\omega_y t) & 0 & \cos(\omega_y t) \end{bmatrix}$$

$$R(t) = R_x(\omega_x t) R_y(\omega_y t)$$

And our angular velocity is defined as

$$\dot{R}(t) R^{-1}(t) = [\boldsymbol{\omega}(t)]_\times$$

Applying the product rule and simplifying gives

$$\dot{R} R^{-1} = \dot{R}_x R_x^{-1} + R_x \left(\dot{R}_y R_y^{-1} \right) R_x^{-1}$$

The first term corresponds to rotation about $\hat{x}$:

$$\boldsymbol{\omega}_x = \begin{bmatrix} \omega_x \\ 0 \\ 0 \end{bmatrix}$$

The term $R_x(\dot{R}_y R_y^{-1}) R_x^{-1}$ represents the ω_y contribution expressed in the current frame, or the vector $(0,\omega_y,0)$ rotated by R_x. Evaluating this transformation,

$$R_x(\omega_x t) \begin{bmatrix} 0 \\ \omega_y \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ \omega_y \cos(\omega_x t) \\ \omega_y \sin(\omega_x t) \end{bmatrix}$$

Thus, the total angular velocity vector is

$$\boxed{\vec{\omega}(t) = \begin{bmatrix} \omega_x \\ \omega_y \cos(\omega_x t) \\ \omega_y \sin(\omega_x t) \end{bmatrix}}$$

Given this angular acceleration vector, we can evaluate our effective acceleration equation, without gravity, as observed in the lab frame, given that $\vec{r} = [x,y,z]$. We obtain the following vector after evaluating symbolically in Wolfram:

$$\vec{a}_{lab}(\vec{r},t) = \begin{bmatrix} -\omega_y^2 x \\ -\left(\omega_x^2 + \omega_y^2 \cos^2(\omega_x t)\right) y - \omega_y^2 \sin(\omega_x t)\cos(\omega_x t) z \\ -\left(\omega_x^2 + \omega_y^2 \sin^2(\omega_x t)\right) z - \omega_y^2 \sin(\omega_x t)\cos(\omega_x t) y \end{bmatrix}$$

3.2 Acceleration as Observed in the Sample Frame

Since we have an equation describing the effective acceleration from inertial contributions as observed in the lab frame, the inverse transform of our rotation matrix is needed to compute the acceleration felt by the sample.

$$\vec{g}_{\text{eff, sample}}(\vec{r},t) = R^{-1}(t) \vec{g}_{\text{lab}}(r,t)$$

Evaluating the transform in wolfram results in our rotated gravity in combination with inertial terms

$$\vec{g}_{\text{eff, sample}}(\vec{r},t) = \begin{bmatrix} -g\cos(\omega_x t)\sin(\omega_y t) \\ g\sin(\omega_x t) \\ g\cos(\omega_x t)\cos(\omega_y t) \end{bmatrix} + \vec{a}_{\text{inertial}}(\vec{r},t)$$

Since we already derived an expression for our inertial terms, we are left with the final equation for the effective “gravitational" acceleration experienced by the rotating sample as a function of time, and displacement from the origin.

$$\boxed{\vec{g}_{\text{eff, sample}}(\vec{r},t) = \begin{bmatrix} -g\cos(\omega_x t)\sin(\omega_y t) \\ g\sin(\omega_x t) \\ g\cos(\omega_x t)\cos(\omega_y t) \end{bmatrix} + \begin{bmatrix} -\omega_y^2 x \\ -(\omega_x^2 + \omega_y^2 \cos^2(\omega_x t)) y - \omega_y^2 \sin(\omega_x t)\cos(\omega_x t) z \\ -(\omega_x^2 + \omega_y^2 \sin^2(\omega_x t)) z - \omega_y^2 \sin(\omega_x t)\cos(\omega_x t) y \end{bmatrix}}$$

4 Effective Gravity as a Function of Time

Using our equation for the effective acceleration felt by the sample, we can examine both quantitatively and qualitatively the quality of microgravity based on varying input parameters.

The characteristic time scale of the system is set by the rotation periods $T_x = \frac{2\pi}{\omega_x}$ and $T_y = \frac{2\pi}{\omega_y}$, which for the chosen parameters correspond to approximately 30 s and 15 s.

We see that the magnitude of our vector stays constant as expected at $9.8\frac{m}{s^2}$. The acceleration for each component is periodic as expected.

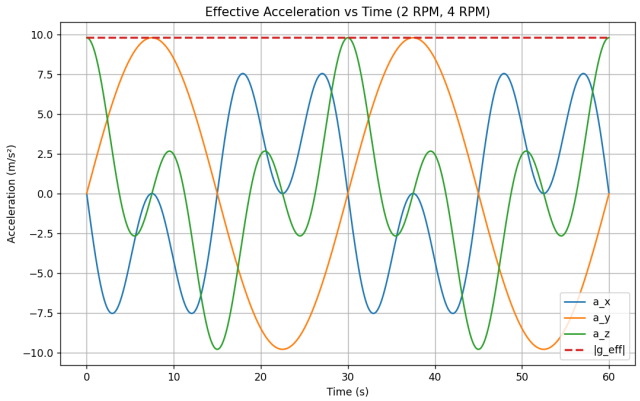


Figure 4.1: Direction and magnitude for effective acceleration given radius and angular speeds

4.1 Verification of Time Averaged Acceleration

The time-averaged effective acceleration over a given time interval is

$$\langle \mathbf{g}_{\text{eff}} \rangle = \frac{1}{T}\int_0^T \mathbf{g}_{\text{eff}}(t)\,dt.$$

This can quickly be computed using Python's `Numpy` library, which we will continuously be using for our computations since the effective time-averaged gravity is used for a majority of the metrics calculated throughout our microgravity analysis sections.

Although we can't always compute the time period for one full cycle, considering certain ratios of angular velocities can result in non-closed paths, we can approximate that one cycle occurs in around 30 seconds. For the speeds of $\omega_x = 2$ rpm and $\omega = 4$ rpm, the magnitude of the time-averaged acceleration vector is

$$\text{Bias} : 0.40288 (m/s^2)$$

This makes sense since we don't expect 30 seconds to fully average out microgravity. We can show the magnitude of the time-averaged acceleration vector a lot more clearly if we graph this average over a larger number of cycles, which just corresponds to a longer period of time.

The following graph, using the same radius and angular velocities, shows how microgravity conditions are effectively replicated over time. For small T we see almost identical behavior to our effective gravity expression. As T gets bigger, our $\frac{1}{T}$ term dominates, and the magnitude of the acceleration components go to zero.

From the running average plot, we observe that the averaged acceleration falls below $\approx 0.01g$ at around $T \approx 500s$ seconds for the chosen parameters. This indicates that effective microgravity conditions require averaging over many rotation cycles, not just a single period.

5 Spatial Analysis of Gravity

Because the effective acceleration depends on position, there is a spatial variation across the sample, which can be described by the gradient $\nabla \mathbf{g}_{\text{eff}}$. This gradient represents how quickly the acceleration changes from one point to another within the

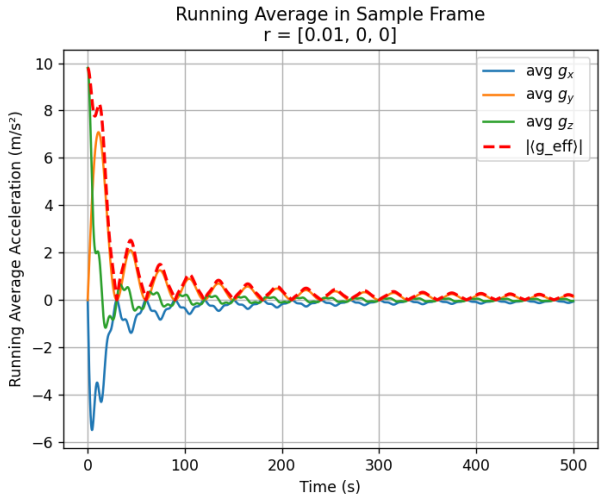


Figure 4.2: Evolution of averaged acceleration vector over time

chamber.

Contour plots follow the key relationship that each contour lies perpendicular to the gradient $\nabla \vec{g}_{\text{eff}}$, meaning we can perform an equivalent spatial analysis without the clutter of a vector plot. The following contour plot is computed using a time average of 20 cycles $\approx 600s$

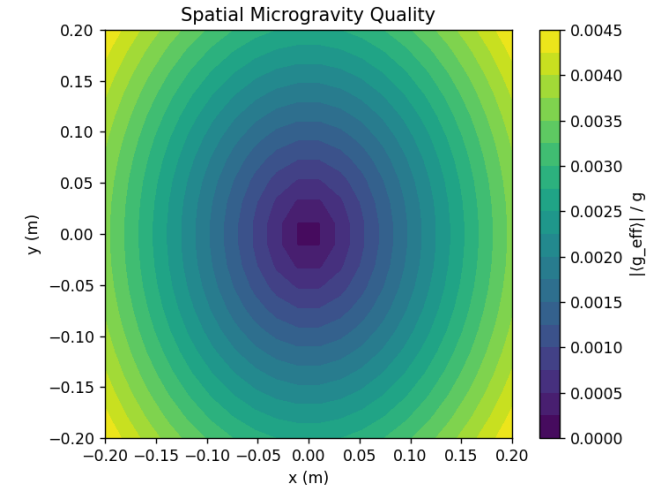


Figure 5.1: Contour Plot of Radial Dependence of Microgravity quality

At the center of rotation ($R = 0$),

$$|\langle \vec{g}_{\text{eff}} \rangle| \approx 0,$$

while at the edge ($R \approx 0.2$ m),

$$|\langle \vec{g}_{\text{eff}} \rangle| \approx 0.0045g \approx 0.044\text{m/s}^2.$$

$$\Delta g_{\text{eff}} \approx 0.0045g \Rightarrow \nabla g_{\text{eff}} \approx \frac{\Delta g}{\Delta R} = \frac{0.0045g}{0.2} \approx 0.0225g/\text{m}.$$

which confirms that the residual acceleration increases roughly linearly with radius. This is expected since our centrifugal force follows a linear relationship with $\vec{r}$.
Cells at opposite sides of the chamber experience similar magnitudes of residual acceleration but different instantaneous directions, while the time-averaged direction remains negligible. This means that maintaining a small to moderate sample size up to 0.10m is ideal to simulate proper microgravity conditions. For larger sample sizes, more cycles are needed to average out microgravity, which can cause potential issues if the overall time is not in line with the time scale of biological processes.

5.1 Sensitivity and Error Analysis

The model is most sensitive to radius and angular speed. The residual acceleration scales approximately as

$$g_{\text{rms,res}} \approx \omega_x \omega_y R,$$

so increasing either the radius or rotation rates increases the residual acceleration. From our results, increasing the radius from 0.05 m to 0.2 m increases the residual acceleration from $\sim 0.0011g$ to $\sim 0.0045g$, showing an approximately linear dependence on R.

If the sample is misaligned from the center of rotation, this introduces error. If the sample is offset by ΔR, the additional acceleration is roughly

$$g_{\text{error}} \sim \omega_x \omega_y \Delta R.$$

For a 0.01 m offset, this is on the order of $10^{-4}g$.

Overall, the maximum variation in the chamber is $\sim 0.0045g$, so minimizing both sample size and misalignment is important for maintaining good microgravity conditions.

6 How Good Is "Microgravity"?

Although gravity is still present, the Clinostat continuously reorients the acceleration vector so that no persistent direction is experienced by the sample. The effectiveness of this approximation can be quantified using the time-averaged bias and RMS acceleration.
From our numerical results, the residual acceleration is

$$|\langle \vec{g}_{\text{eff}} \rangle| \approx 0.041g \text{ or } 0.40\text{m/s}^2,$$

indicating that a small but nonzero directional bias remains. In contrast, the RMS acceleration satisfies

$$g_{\text{rms}} \approx 1.00g,$$

showing that the magnitude of the acceleration is not reduced, but instead varies in direction over time.
The characteristic rotation period of the system is

$$T_{\text{rot}} \approx 30\text{s},$$

while the biological response time for *Tetrahymena* is on the order of

$$\tau_{\text{bio}} \sim 7200\text{s}.$$

This gives

$$\frac{T_{\text{rot}}}{\tau_{\text{bio}}} \approx 4 \times 10^{-3},$$

indicating that the direction of the effective acceleration changes much faster than the organism can respond.
Therefore, although the magnitude of gravity remains unchanged, its direction is rapidly randomized. The residual bias is roughly negligible ($\sim 4\%$ of g). Since the rotation occurs on a time scale much shorter than the biological response we can increase the amount of cycles we average over. Averaging over two hours, we get

$$\boxed{|\langle \vec{g}_{\text{eff}} \rangle| \approx 0.00176\text{m/s}^2 \approx 1.79 \times 10^{-4} g.}$$

Which means our sample is undergoing concrete microgravity conditions over the respective biological time frame. Together, these results demonstrate that the Clinostat effectively suppresses sustained directional loading and produces a microgravity-like environment.

7 Biological Interpretation

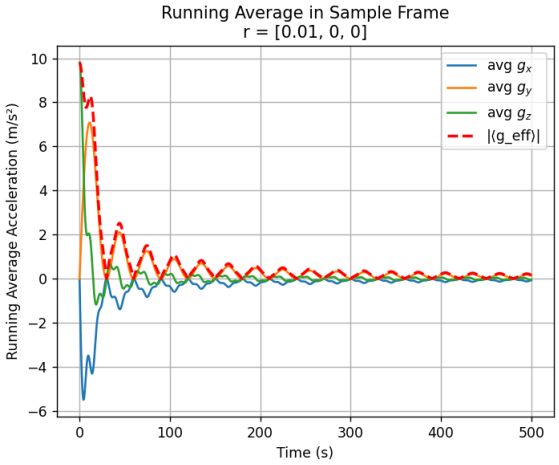
7.1 What are the cells actually experiencing?

The purpose of a clinostat is not to remove gravity, but to remove the ability of cells to sense a constant direction of gravity. Under normal conditions on Earth, gravity provides a persistent downward cue that many biological systems rely on for orientation, transport processes, and signaling. In a clinostat, this directional cue is continuously changed by rotating the sample about one or more axes.

As a result, the direction of the effective gravitational force acting on the cells is constantly changing. From the perspective of a cell, there is no longer a stable "down" direction. Instead, the acceleration vector rotates over time, meaning that any internal biological processes that depend on a fixed gravitational direction are disrupted. Over sufficiently long times, this produces an effect similar to microgravity, not because gravity is eliminated, but because its direction is no longer consistent.

However, this cancellation is not perfect. Only a point located exactly at the center of rotation experiences a nearly complete averaging of gravitational direction. Cells located away from the center experience additional accelerations due to the rotation of the system. These accelerations introduce what is known as *residual acceleration*, meaning that the environment is not truly weightless. Instead, cells experience a small, time-varying acceleration whose magnitude depends on their distance from the center.

For example, in a typical setup with a maximum radius of approximately 13.5 cm, the residual acceleration at the outer edge can reach a small but non-negligible fraction of Earth's gravity. While this is much smaller than the full gravitational acceleration, it is still present and may influence sensitive biological processes.



The figure above illustrates how the time-averaged acceleration at the center approaches zero, demonstrating that the clinostat is most effective when the sample is positioned as close to the center as possible.

7.2 How effective is clinostat-simulated microgravity?

The effectiveness of a clinostat depends on two key factors: how quickly the direction of gravity changes, and how large the residual accelerations are within the sample.

First, the rotation speed determines how rapidly the direction of the gravitational vector is reoriented. If the system rotates quickly compared to the time scale over which cells respond to mechanical stimuli, then the cells effectively experience an average of all possible orientations. Many biological responses to gravity occur on time scales of seconds to minutes. In contrast, a Clinostat rotating with a period of a few seconds changes the direction of gravity much faster than the cells can respond. Under these conditions, the cells are unable to detect a consistent direction, and the environment behaves similarly to microgravity.

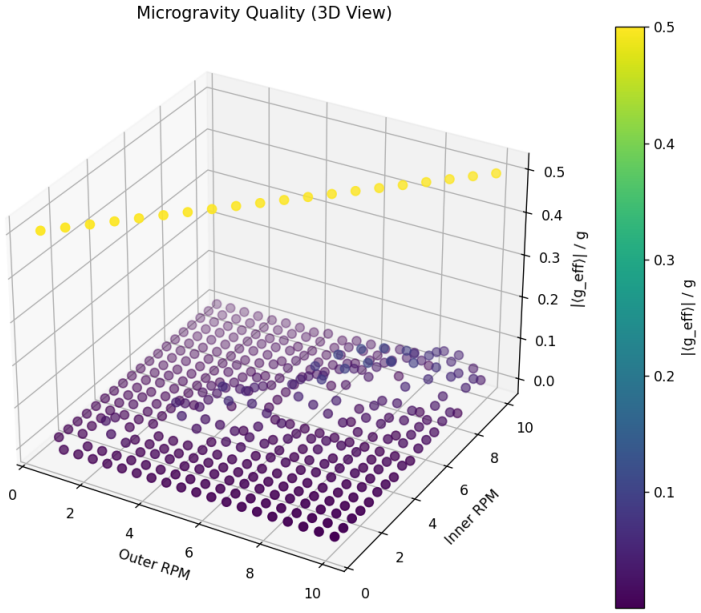


Figure 7.1: Quality of Microgravity at different RPMs averaged over 200s

This graph shows how the quality of simulated microgravity depends on the rotation speeds of the clinostat's two axes. Each point represents a different combination of inner and outer rotation rates, and the height/color indicates how much residual acceleration remains compared to Earth's gravity. Lower values denote better microgravity conditions, while higher values mean cells are still experiencing non-negligible forces. The main takeaway is that microgravity is not a given. Only certain rotation settings effectively minimize acceleration. If the system is not operated in this optimal range, cells will sense a directional force and behave more like they would under normal gravity.

The choice of rotation rates also plays an important role. Using two rotation axes with non-integer ratios (for example, 1:1.6) helps ensure that the system does not repeat the same orientations over time. This leads to a more uniform sampling of all possible directions and improves the averaging effect. Ratios such as 1:π provide an even broader distribution of orientations, but may introduce practical challenges in implementation. In general, avoiding simple integer ratios is important for preventing periodic motion that could reintroduce directional bias.

Second, the size and positioning of the sample determine the magnitude of residual accelerations. Cells located farther from the center experience larger rotational effects, which increases the residual acceleration they feel. This means that even though the direction of gravity is changing, the magnitude of the acceleration is not negligible everywhere in the sample. As a result, larger samples or samples that are not well-centered will experience less effective microgravity conditions.

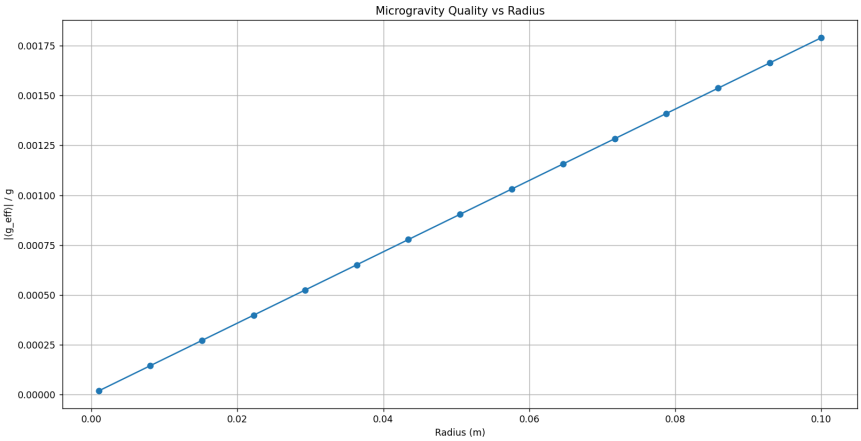


Figure 7.2: Quality of Microgravity at different radii

This graph shows how the residual acceleration increases with distance from the center of the chamber. The nearly linear trend means that cells at the center experience very low acceleration (closest to microgravity), while cells farther out experience progressively stronger forces. Biologically, this is important because it means not all cells in the sample experience the same environment. Cells near the edges may respond differently than those near the center, introducing variability into experiments and making sample size and positioning critical factors in achieving reliable results.

Despite these advantages, clinostats have important limitations. The presence of fluid within the sample can introduce additional forces, such as shear stresses and internal flow, which are not accounted for in a simple rigid-body model. These effects can influence how cells experience their environment and may lead to deviations from ideal microgravity conditions. Furthermore, residual accelerations cannot be completely eliminated except at a single point, meaning that perfect microgravity is never achieved.

7.3 When does the clinostat work well?

A clinostat provides a good approximation of microgravity under the following conditions:

- The sample is positioned very close to the center of rotation or roughly less than 10 centimeters.
- The overall size (radius) of the sample is small. The current sample container dimensions are adequate for microgravity expirements.
- The rotation rate is fast compared to biological response times.
- The rotation axes use non-repeating (non-integer) ratios.

When these conditions are satisfied, the residual acceleration experienced by the cells can be reduced to a small fraction of Earth's gravity, and the constantly changing direction prevents the cells from sensing a stable gravitational cue.

However, if the sample is large, poorly centered, or rotated too slowly, the approximation breaks down. In these cases, cells may still experience a noticeable directional bias or sufficient acceleration to influence biological behavior.

7.4 Summary for biological applications

In summary, a clinostat does not eliminate gravity, but instead makes it difficult for cells to detect a consistent direction of gravitational force. This creates an environment that mimics key aspects of microgravity, particularly the absence of a stable "down" direction. The effectiveness of this simulation depends strongly on rotation speed, sample size, and positioning.

For experimental applications, this means that careful design is essential. Samples should be kept small and centered, and rotation speeds should be chosen to ensure rapid reorientation without introducing excessive mechanical stress. When used under these conditions, a clinostat can serve as a powerful tool for studying how biological systems behave in environments where directional gravity cues are absent.